

SEQUENTIAL PROCEDURES BASED ON KENDALL'S TAU STATISTIC

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SUMMARY

We consider the linear regression model $X_i = \alpha + \beta t_i + \epsilon_i$, $i = 1, 2, \dots$ where the ϵ_i are i.i.d.r.v. with unknown d.f. F , and where it is assumed that $t_1 < t_2 < \dots$. Sequential inference on β is considered, based on Kendall's Tau statistic U_n defined in (1.1) below.

We show that $\binom{n}{2} U_n$ is a sum of independent r.v. with known but not identical distributions. From this the law of the iterated logarithm for U_n is derived and a nonparametric confidence sequence found for β . Also, we derive two invariance principles for U_n and these are used to study a truncated sequential test of $H_0: \beta = 0$ vs. $H_1: \beta > 0$.

1. Introduction

Let $\{X_i\}$ be a sequence of independent r.v.'s with absolutely continuous c.d.f.'s $\{F(x - \alpha - \beta t_i)\}$ where $\{t_i\}$ is a sequence of known constants and α, β are unknown. A distribution free test of $H_0: \beta = 0$ can be based on the statistic

$$(1.1) \quad U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} S(t_j - t_i) S(X_j - X_i)$$

where $S(u) = 2\mathbf{1}(u) - 1$ and $\mathbf{1}$ is the indicator of set $\{u > 0\}$. Here we investigate some sequential procedures based on U_n , under the simplifying assumption

$$(1.2) \quad t_1 < t_2 < t_3 < \dots$$

Our results are stated in Section 2, while Section 3 contains the proofs.

It should be noted that sequential procedures can be based on a general class of regression rank statistics. Our main reason for considering (1.1) instead is the comparative ease of calculation of that statistic and of the associated confidence interval for β .

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Throughout, P_a is a probability measure under which the X_i are independent with absolutely continuous c.d.f. $F(x - \alpha - at_i)$, while P^* is a probability measure under which the process $(B(t), 0 \leq t \leq 1)$ is standard Wiener. For the notions of weak convergence in the space $C[0,1]$ used in this paper, we refer the reader to Gikhman and Skorokhod (1969), Chapter 9.

2. Statement of Results

Assuming (1.2), we can write (1.1) as

$$\begin{aligned} (2.1) \quad \left(\frac{n}{2}\right) U_n &= 2 \sum_{j=2}^n \left(\sum_{i=1}^{j-1} 1(X_j - X_i) + 1 \right) - \frac{(n-1)(n+4)}{2} \\ &\equiv 2 \sum_{j=2}^n Y_j - \frac{(n-1)(n+4)}{2} \\ &\equiv 2 S_n - \frac{(n-1)(n+4)}{2} \end{aligned}$$

The relative ease with which the results of this section are derived is principally due to the following theorem.

Theorem 2.1

(i) $Y_2, Y_3, \dots$ are P_0 -independent.

(ii) For $n \geq 2$: $P_0 \{Y_n = i\} = n^{-1}$, $i = 1, \dots, n$.

(iii) $E_0 S_n = (n-1)(n+4)/4$;

$$\text{var}_0 S_n \equiv V_n = n(n-1)(2n+5)/72.$$

(iv) $E_0 \{ \exp(t(S_n - E_0 S_n)) \} \leq \exp(nt/2 + t^2(n^3/3 + n^2/2 + n/6)/24)$ for $t > 0$.

Consider the problem of constructing a confidence sequence for β . Let $\beta_{L,n} < \beta_{U,n}$, $n = r, r+1, \dots$ denote two sequences of r.v.'s (with $\beta_{L,n}$ and $\beta_{U,n}$ each measurable only on $X_1, \dots, X_n$) such that for prescribed $0 < \alpha < 1$.

$$(2.2) \quad P_\beta \{ \beta \in (\beta_{L,n}, \beta_{U,n}) \text{ for all } n \geq r \} \geq 1 - \alpha.$$

Then $\{(\beta_{L,n}, \beta_{U,n}); n \geq r\}$ is said to constitute a $1 - \alpha$ confidence sequence for β . It follows immediately from (2.2) that if n^* is any stopping rule of the sequence $X_1, X_2, \dots$,

then

$$(2.3) P_{\beta} \{ \beta \in (\beta_{L,n*}, \beta_{U,n*}) \} \geq 1 - \alpha$$

In order to construct a confidence sequence for β , let $c(n, \alpha)$, $n \geq r$ be a sequence of constants such that

$$(2.4) P_0 \{ U_n \geq +c(n, \alpha), \text{ some } n \geq r \} \leq \alpha$$

and let

$$(2.5) a(n, \alpha) = \min \left\{ \binom{n}{2}; \left[\frac{1}{2} \cdot \binom{n}{2} (1 + c(n, \alpha)) + 1 \right] \right\}$$

$$b(n, \alpha) = \max \left\{ 1; \left[\frac{1}{2} \cdot \binom{n}{2} (1 - c(n, \alpha)) - 1 \right] \right\}.$$

Denoting by $X_{n,1} < \dots < X_{n,n(n-1)/2}$ the order statistics of the set of divided differences $\{(X_j - X_i)/(t_j - t_i); 1 \leq i < j \leq n\}$ it has been shown by Steyn (1973), pp. 132-134, that it suffices to take

$$(2.6) \beta_{L,n} = X_{n,b(n,\alpha)} \text{ and } \beta_{U,n} = X_{n,a(n,\alpha)}.$$

Returning to (2.4), it is obvious from (2.5) and (2.6) that in order for $\beta_{U,n} - \beta_{L,n}$ to be as small as possible at each $n \geq r$, $c(n, \alpha)$ should decrease as rapidly as possible. Steyn (1973) obtained $c(n, \alpha) \sim B(\log n/n)^{1/2}$ with $B > 2$. The following result shows that one should be able to do even better asymptotically

Lemma 2.1

$$\overline{\lim}_{n \rightarrow \infty} \frac{n^{1/2} U_n}{(8 \log_2 n/9)^{1/2}} = \lim_{n \rightarrow \infty} \frac{n^{1/2} U_n}{(8 \log_2 n/9)^{1/2}} = 1$$

where $\log_2 n = \log \log n$.

This indicates that one should strive for

$$(2.7) c(n, \alpha) \sim (8 \log_2 n/9n)^{1/2} \text{ as } n \rightarrow \infty.$$

Theorem 2.2

Let $K_j = \sum_{i=j}^{\infty} i^{-1} (\log i)^{-2}$ where j is such that $r = [\exp(j/\log j)]$.

Then

$$c(n, \alpha) = 2(n-1)^{-1} + \{8/9(n^{-1/2} + 3n^{-3/2}/2 + n^{-5/2}/2) \\ \cdot (\log_2 n + 4 \log_3 n + 3 \log_2 + \log K_j - \log \alpha)\}^{1/2} \\ n(n-1)^{-1} \exp(3/2 \log_2 n)$$

satisfies (2.4) and (2.7).

We now consider the asymptotic distribution of U_n . Asymptotic normality of U_n under P_0 may be proved by invoking the theory of U-statistics, since it is clear that U_n is a U-statistic in the independent but non-identically distributed random vectors (t_i, X_i) , $i = 1, 2, \dots, n$. However, a routine calculation shows that $Y_2, Y_3, \dots$ satisfy the Lindeberg condition, hence the convergence of $\frac{3}{2} n^{1/2} U_n$ under P_0 to the standard normal distribution. We can get an even stronger result. Let $s_0^2 = 0$ and $s_k^2 = \frac{2}{9} \frac{k+1}{k} (2k+7)$ for $k \geq 1$.

Theorem 2.3

Let $\{X_m(t), 0 \leq t \leq 1\}$ be the polygonal random function in $C[0,1]$ obtained by linear interpolation between the points $(\frac{3}{2}(k+1) U_{k+1}/m^{1/2}, s_k^2/s_m^2)$, $k = 0, 1, \dots, m$ where $U_1 \equiv 0$.

Then, under P_0 , $X_m(\cdot)$ weakly converges to a standard Wiener process on $[0,1]$ as $m \rightarrow \infty$.

If we work' under P_γ , $\gamma \neq 0$ then Theorem 2.1 ceases to be valid. A type of invariance principle can nevertheless be obtained upon use of Theorem 3.1 of Ghosh and Sen (1971). In the following theorem we use the notation, and make the assumptions (i) – (v) of Ghosh and Sen (1971), p. 26. $U_n(a)$ denotes the statistic U_n calculated on the shifted sequence $X_i - a_i$, $i = 1, 2, \dots$. We also put $B(F) = \int f^2(x) dx$ where f is the density function of F .

Theorem 2.4

Let $\{Y_m(t), 0 \leq t \leq 1\}$ be the polygonal random function of $C[0,1]$ obtained by linear interpolation between the points

$$(\frac{3}{2}(k+1) U_{k+1} (-\gamma T_{m+1}^{-1})/m^{1/2} - \gamma \sqrt{12} B(F) \rho(\frac{k+1}{m}) \cdot s(\frac{k+1}{m})^{1/2}, s_k^2/s_m^2),$$

$k = 0, 1, \dots, m$; γ a constant.

Then, under P_0 , $Y_m(t)$ weakly converges to a standard Wiener process as $m \rightarrow \infty$.

Theorem 2.5

Both Theorems 2.3 and 2.4 remain valid if s_k^2/s_m^2 is replaced by k/m .

Finally we consider a truncated sequential test of $H_0: \beta = 0$ v.s. $H_1: \beta > 0$. Using asymptotic likelihood ratio considerations based on Theorem 2.4, in a similar way as Miller (1970), p. 100, we arrive at the following:

Let α , $0 < \alpha < 1$ be specified and let $a(\alpha)$ be the upper α -point of the standard normal distribution. Let also the truncation point $m + 1$, an integer ≥ 2 , be specified in advance. The sequential procedure works as follows:

Accept $H_0: \beta = 0$ if $\frac{3}{2}k U_k < a(\alpha) \sqrt{m}$ for all $k = 2, \dots, m + 1$,

Accept $H_1: \beta > 0$ if $\frac{3}{2}k U_k \geq a(\alpha) \sqrt{m}$

The asymptotic properties as $m \rightarrow \infty$ (e.g. power, sample size distribution) now follow in a straightforward way from Theorems 2.3, 2.4 and 2.5. In particular,

$$\begin{aligned} \lim_{m \rightarrow \infty} P_0(\text{reject } H_0) &= P^*(B(t) > a(\alpha) \text{ for some } 0 \leq t \leq 1) \\ &= 2 P^*(B(1) > a(\alpha)) \\ &= \alpha \end{aligned}$$

and for $\gamma > 0$:

$$(2.8) \quad \lim_{m \rightarrow \infty} P_{\gamma T_m - 1}(\text{accept } H_1) = P^*(B(t) > a(\alpha) - \sqrt{1/2} B(F)\rho(ts(t)))^{1/2} \text{ for some } 0 \leq t \leq 1).$$

Thus, as the truncation point $m + 1 \rightarrow \infty$, the level of significance approaches the specified value α , while the power function tends to a quantity that depends on a particular regression design used.

3. Proofs of the Theorems

For the proof of Theorem 2.1, note that Y_k is the rank of X_k in the sequence $X_1, \dots, X_k$. Thus (i) and (ii) follow from Theorem 1.1 of Barndorf-Nielsen (1963).

Part (iii) follows by direct calculation.

As to (iv), note that for $t > 0$,

$$\begin{aligned} 1/k \sum_{i=1}^k \exp \left\{ t \left(i - \frac{k+1}{2} \right) \right\} &\leq 1/k \int_1^{k+1} \exp \left\{ t \left(x - \frac{k+1}{2} \right) \right\} \cdot dx \\ &= 1/kt (\exp(kt/2) - \exp(-kt/2)) \cdot \exp(t/2) \\ &\leq \exp(t/2 + t^2 k^2 / 24) \cdot \end{aligned}$$

Straightforward calculations based on parts (i) – (iii) of Theorem 2.1 show that Lemma 2.1 is a direct consequence of Kolmogorov's law of the iterated logarithm – See Loève (1963), p. 260.

Coming to Theorem 2.2, note that if $S_n^* = S_n - E_0 S_n - n/2$, then

$\{\exp(t S_n^*) / E_0 \exp(t S_n^*); n \geq 2\}$ is a P_0 – martingale. Letting $m_i (\rightarrow \infty)$ be any increasing sequence of positive constants and $t_i > 0$ any sequence of constants, and using Theorem 2.1 (iv), we obtain in similar fashion as Darling & Robbins (1967), p. 1189,

$$P_0 \{S_n^* > (n^3/3 + n^2/2 + n/6)/12 \cdot b_n \text{ for some } n \geq m_j\} \\ \leq \sum_{i=j}^{\infty} \exp \{-(m_i^3/3 + m_i^2/2 + m_i/6)t_i^2/24\} \equiv A_j$$

where $b_n = t_i$ for $m_i \leq n < m_{i+1}$, $i = 1, 2, \dots$

If $m_i = \exp(i/\log i)$ and

$$t_i = (24 \log i + 48 \log_2 i + 24 \log K_j - 24 \log \alpha)^{1/2} (m_i^3/3 + m_i^2/2 + m_i/6)^{-1/2} \\ < (24 \log i + 48 \log_2 i + 24 \log K_j - 24 \log \alpha)^{1/2} (3/m_i^3)^{1/2},$$

then it follows that $A_j = \alpha$, and from the inequalities at the bottom of page 1189 of Darling and Robbins (1967) that

$$t_i < \{(24 \log_2 n + 96 \log_3 n + 72 \log 2 + 24 \log K_j - 24 \log \alpha) \cdot 3n^{-3} \cdot \exp \\ (3/\log_2 n)\}^{1/2}$$

To complete the proof, we need only note that $(\frac{n}{2}) U_n = 2(S_n - E_0 S_n)$. To prove Theorem 2.3, let

$$Z_{i-1,n} = \frac{4Y_i}{n} - \frac{(n-1)(n+4)}{n} \quad i = 2, 3, \dots, n+1 \text{ and } Z_n = \sum_{j=1}^n Z_{j,n}, n = 1, 2, \dots \text{ with } Z_0 \equiv 0.$$

Note that $Z_n = (n+1) U_{n+1}$

From Theorem 2.1 we get

$$E Z_i = 0, \text{ var } Z_n = S_n^2 = \text{var} \frac{4S_{n+1}}{n} = \frac{2}{9} \frac{n+1}{n} (2n+7) \sim \frac{4}{9} n.$$

Let $(Z_m(t), 0 \leq t \leq 1)$ be the polygonal random function in $C[0,1]$ obtained by linear interpolation between points $(\frac{3}{2} Z_k/m^{1/2}, s_k^2/s_m^2)$ $k = 0, 1, \dots, m$ with $s_0^2 \equiv 0$.

Then by Theorem 1, page 452 of Gikhman and Skorokhod (1969) we have that $Z_m(\cdot)$ weakly converges to a standard Wiener process.

Since $Z_m(\cdot) = X_m(\cdot)$, Theorem 2.3 follows.

For the proof of Theorem 2.4, let

$$W_k(a) = k U_k(a/\rho_k T_k) - k U_k + 8a B(F)(k(k+1)/12(k-1))^{1/2}$$

and

$$V_k = \sup \{|W_k(a)| : |a| < \log k\}$$

Now write, with $X_m(\cdot)$ defined in Theorem 2.3

$$\begin{aligned} (2.9) \quad & Y_m(S_k^2/S_m^2) - X_m(S_k^2/S_m^2) \\ &= \frac{3}{2}(k+1) U_{k+1}(-\gamma/T_{m+1})/m^{1/2} - \gamma\sqrt{12} B(F) \rho \left(\frac{k+1}{m}\right)^{1/2} - \frac{3}{2}(k+1) U_{k+1}/m^{1/2} \\ &= \frac{3}{2} W_{k+1}(-\gamma\rho_{k+1} T_{k+1}/T_{m+1})/m^{1/2} + \gamma\sqrt{12} \cdot B(F) \cdot (k+1)^{1/2} \cdot \frac{T_{k+1}}{T_{m+1}} \\ &\quad \{ \rho_{k+1}(k+2)^{1/2}/k^{1/2} - \rho \}/m^{1/2} + \gamma\sqrt{12} B(F) \cdot (k+1)^{1/2} \cdot \rho \left\{ \frac{T_{k+1}}{T_{m+1}} - s^{1/2} \left(\frac{k+1}{m}\right) \right\}/m^{1/2} \\ &= \frac{3}{2} W_{k+1}(-\gamma\rho_{k+1} T_{k+1}/T_{m+1})/m^{1/2} + a_k/m^{1/2} \cdot T_{m+1} + b_{k,m}/m^{1/2}. \end{aligned}$$

Since $\rho_k \rightarrow \rho > 0$ and $T_k \uparrow \infty$, it follows that

$$|\gamma|\rho_{k+1} T_{k+1}/T_m < \log(k+1) \text{ for } k = k_0, k_0 + 1, \dots, m,$$

$$\text{where } k_0 = [\exp \{ |\gamma| \sup_{k \geq 2} \rho_k \}]$$

Hence

$$\begin{aligned} (2.10) \quad & \max_{0 \leq k \leq m} |W_{k+1}(-\gamma\rho_{k+1} T_{k+1}/T_{m+1})| \\ & \leq \max_{0 \leq k \leq k_0-1} |W_{k+1}(-\gamma\rho_{k+1} T_{k+1}/T_{m+1})| + \max_{k_0 \leq k \leq m} V_k \\ & = o(m^{1/2}) \text{ a.s. } P_0, \end{aligned}$$

since by Theorem 3.1 of Ghosh and Sen (1971) it holds that $V_k = o(k^{1/2})$ a.s. P_0

Also, since $\rho_k \frac{(k+2)^{1/2}}{k^{1/2}} \rightarrow \rho$, we find that $a_k = o((k+1)^{1/2} T_{k+1})$

Consequently

$$(2.11) \quad \max_{0 \leq k \leq m} |a_k| = o(m^{1/2} T_{m+1})$$

Finally, from assumption (v) on page 26 of Ghosh and Sen (1971)

$$(2.12) \quad m^{-1/2} \max_{0 \leq k \leq m} |b_{k,m}| \leq \left(\frac{m+1}{m} \right)^{1/2} \max_{0 \leq k \leq m} \left| \frac{T_{k+1}}{T_{m+1}} - s^{1/2} \left(\frac{k+1}{m} \right) \right| \cdot \gamma \sqrt{12} B(F) \rho$$

$$\leq \left(\frac{m+1}{m} \right)^{1/2} \left\{ \sup_{0 \leq t \leq 1} \left| \frac{Q([mt] + 1)}{Q(m+1)} - s^{1/2}(t) \right| \right\}$$

$$+ \sup_{0 \leq t \leq 1} |s^{1/2} \left(\frac{[mt] + 1}{m} \right) - s^{1/2}(t)| \cdot \gamma \sqrt{12} B(F) \rho$$

$$\rightarrow 0 \text{ as } m \rightarrow \infty$$

Since $\sup_{0 \leq t \leq 1} |X_m(t) - Y_m(t)| = \max_{0 \leq k \leq m} |X_m(s_k^2/s_m^2) - Y_m(s_k^2/s_m^2)| \rightarrow 0$ a.s. P_0 ,

by (2.9)–(2.12), it follows that $Y_m(\cdot)$ has the same limit distribution as $X_m(\cdot)$. The proof is complete.

Finally, a routine computation shows that there is a constant C such that

$$\max_{0 \leq k \leq m} \left| \frac{s_k^2}{s_m^2} - \frac{k}{m} \right| \leq C m^{-1}$$

for all m sufficiently large. The proof of Theorem 2.5 thus follows from the tightness of the sequence $\{X_m(t), 0 \leq t \leq 1\}$ along the lines of Sen (1974), p. 57.

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